

Samba Hama TRAORE

DATA ENGINEER

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PROFILE

Final-year Computer Science student (Big Data / AI) at Dakar Institute of Technology, specializing in data engineering: ingestion, transformation, modeling, and pipeline orchestration. A hands-on program requiring an annual capstone project and defense, complemented by end-to-end personal projects documented and versioned on GitHub. Creative, curious, and detail-oriented; seeking to contribute to data teams.

TECHNICAL SKILLS

LANGUAGES	Python (pandas, NumPy, PySpark), SQL, R, Scala, JavaScript
BIG DATA & ORCHESTRATION	Spark, Kafka, Hadoop, Airflow, dbt
DATABASES	PostgreSQL, MySQL, MongoDB (NoSQL)
CLOUD & DEVOPS	AWS, Google Cloud, Docker, Git, GitHub Actions, Jenkins
FRAMEWORKS & WEB	Flask, FastAPI, Vue.js, React, Streamlit, Tailwind, HTML5 / CSS3
DATA VIZ & MONITORING	Dash, Power BI, Matplotlib, Seaborn, Tableau, Grafana, Prometheus
TOOLS	Linux / Bash, VS Code, Jupyter, Jira, Trello, Selenium, Markdown

PROJECTS

• CloudIT : Distributed High-Performance Computing Platform on Multi-GPU Infrastructure

Browser-based service that runs data processing, AI training, and other compute-intensive jobs on shared GPU servers, with automatic workload distribution and per-user quotas.

PYTHON · REACT/TYPESCRIPT · POSTGRESQL · REDPANDA · DOCKER · MINIO · JWT

• Data Processing : Multi-Sector Dataset Cleaning & Analytics

Processed and cleaned 12 datasets across 9 sectors : health, epidemiology, employment, education, environment, transportation, retail, rentals, customer relations; with Seaborn and Plotly visualizations and an interactive Dash dashboard.

PYTHON · PANDAS · SEABORN · PLOTLY · DASH

• FAR_sMart : Automated Monitoring of 4 Connected Agricultural Fields

Streaming platform that continuously collects and processes field conditions from on-site robots, surfacing results in a dashboard built for farmers.

PYTHON · APACHE KAFKA · APACHE SPARK · APACHE AIRFLOW · MINIO · PARQUET · DOCKER

• Weath_Air : Automated Weather Data Collection Pipeline

Hourly pipeline that retrieves, transforms, and archives weather readings with no manual intervention and automatic retry on service failure.

PYTHON · APACHE AIRFLOW · REST API

• Sparkineering : Distributed Analysis of a Real-Estate Dataset

Analysis of San Francisco Airbnb listings with Spark, surfacing price trends, neighborhood rankings, and correlations between price and guest satisfaction.

PYSPARK · SPARK SQL

• CardioFlow : Heart Disease Risk Prediction

End-to-end ML workflow from model training to API deployment, with a Streamlit interface for prediction and performance monitoring.

PYTHON · SCIKIT-LEARN · MLFLOW · FASTAPI · STREAMLIT · DOCKER · GITHUB ACTIONS

EDUCATION

• B.S. in Computer Science, Big Data / AI

Dakar Institute of Technology (DIT), Dakar, Senegal

Oct 2024 - Present

• Computer Science coursework (networking, fundamentals)

ITMA and Faculty of Science and Technology (FST), Bamako, Mali

2021 - 2023

• High School Diploma, Science track

École Maarif de Turquie, Bamako, Mali

Oct 2018 - Jun 2021

LEADERSHIP

President, Student Union - DIT

Led clubs and tech workshops, a blood drive, meal distributions, and community events.

President, External Relations Committee - DIT

Built partnerships with tech organizations: training sessions, guest speakers, company visits.

Class Representative, Years 1-3 - DIT

Liaison between the class and faculty; coordinated group projects.

CERTIFICATIONS & LANGUAGES

AWS Educate

Getting Started with Databases · Storage · Security · Networking

Languages

French (fluent) · English (professional working proficiency) · Bambara (native)